

Technical Report - **Product specification**

EasySpot

Course: IES - Introdução à Engenharia de Software

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Project Abstract: EasySpot is a real-time parking analytics platform that shows live parking occupancy, EV charger status, and accessible-bay availability across urban sites.

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1. Introduction

EasySpot is a parking analytics app that provides real-time parking occupancy, EV charger status, and accessible bay availability. Drivers can quickly view available bays and fixed hourly rates on the app. Operators access read-only dashboards with occupancy and revenue reports, while technicians monitor sensor health, diagnostics, and event logs.

2. Product concept

Vision Statement

Concept

EasySpot, is a smart parking management platform that provides real-time location and monitoring of parking spaces. It offers a platform across web and mobile interfaces for all users: a driver experience to find, filter, and navigate to available spots, and a data dashboard for operators. The system integrates IoT sensor APIs to capture occupancy data, generate usage statistics, and provide predictive insights.

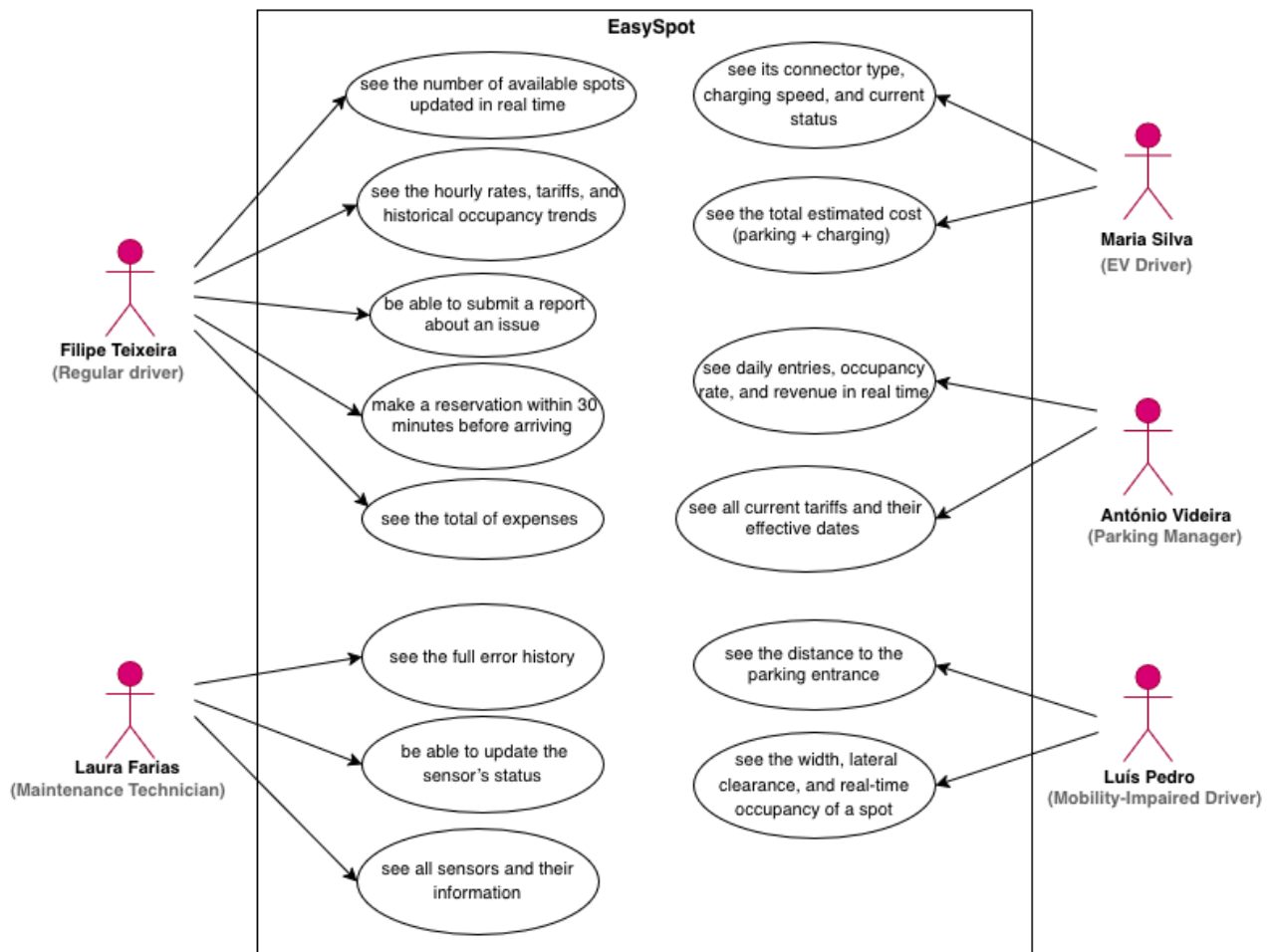
Inspiration

While there are existing navigation tools like Google Maps or Waze that offer basic location services, they often lack granular, real-time data regarding specific parking availability and specialized needs (such as EV charging status or accessibility features). We were inspired by the growing frustration of "circling the block" in urban centers, which wastes time. Unlike traditional systems, EasySpot differentiates itself by offering routing for electric vehicles and disabled drivers, alongside a management API that allows parking owners to monitor financial returns and hardware health in one place.

Motivation

The motivation behind EasySpot comes from the need to optimize urban space and improve the daily lives of diverse user groups. We wanted to solve the "last-mile" anxiety, the uncertainty of whether a spot will be available upon arrival. By providing real-time data on charger compatibility for EV owners and guaranteed accessibility information for users with reduced mobility, we aim to make city navigation more inclusive. Additionally, we are driven to give parking managers better tools to monitor their infrastructure, moving away from outdated, non-data-driven management methods.

Use Case Diagram



Personas

Persona #1: The regular driver

Name: Filipe Teixeira

Age: 25

Location: Coimbra, Portugal

Occupation: Economist



Description: Filipe is a young economist from Coimbra who has to frequently drive to Lisbon for weekly meetings with clients and partners. For Filipe, time is his most valuable resource, and he views parking not just as a necessity but as a logistical challenge that needs to be optimized to ensure his punctuality and professional image. If he feels that he might not be able to get a spot on time, he can make a reservation within 30 minutes before arriving at the parking lot.

Motivation: Wants to minimize search time. He needs to check parking availability in advance to avoid driving in circles, ensuring he arrives at his meetings exactly on time without the stress of being late.

- Appreciates price-to-distance ratios: As an economist, he naturally compares costs. He wants to find the best balance between a low-cost spot and its proximity to his specific meeting location.
- Needs transparent and real-time data: He is frustrated by confusing rate structures and outdated information. He expects the app to show the exact price and a live status of the spots to make quick, data-driven decisions.

Persona #2 : The EV (Electric Vehicle) Driver

Name: Maria Silva

Age: 30

Location: Torres Vedras, Portugal

Occupation: Veterinarian



Description: Maria is a veterinarian living in Torres Vedras who commutes daily to her clinic in the city center. Driving a Tesla, her commute isn't just about the 30-minute journey, but also about ensuring her vehicle remains functional for her busy schedule. For Maria, a parking spot must be more than just a space; it must be a reliable service that supports her commitment to electric mobility while fitting into her demanding professional routine.

Motivation:

- **Seeks guaranteed charging infrastructure:** Maria's primary motivation is to eliminate her anxiety by finding parking spots equipped with functional EV chargers. She needs to ensure her car is charging while she works so it is ready for her return trip.
- **Values punctuality and proximity:** As a veterinarian with a tight schedule, she is motivated to find spots close to her office. This avoids long walks and ensures she arrives on time for her job, including during peak hours.
- **Demands technical transparency:** She is frustrated by the lack of specific information regarding charger types, charging speed, prices, and real-time availability. She expects the app to provide live status updates on chargers to avoid arriving at a station only to find it occupied or out of service, allowing her to make efficient, stress-free decisions.

Persona #3 : The Mobility-Impaired Driver

Name: Luís Pedro

Age: 70

Location: Faro, Portugal

Occupation: Retired



Description: Luís has reduced mobility and uses a wheelchair. He frequently visits clinics and hospitals and needs to ensure that his parking spot is not only available but also practical for his specific physical needs.

Motivation: To maintain independence by finding parking spots that are close to his destination (medical services) and that provide enough space to maneuver his wheelchair safely.

- Appreciates: Accurate real-time information about accessible spots, clear walking (or rolling) distances to entrances, and an easy-to-use interface with high contrast.
- Needs: A way to filter specifically for "Accessible/Disabled" spots; To see the exact distance between the spot and the building entrance; Information on whether spots are being monitored to prevent unauthorized use; Assurance that the spot has the necessary lateral space for a wheelchair ramp or exit.

Persona #4 : The Parking Manager

Name: António Videira

Age: 50

Location: Leiria, Portugal

Occupation: Parking Manager



Description: António manages several parking sites, including the one integrated with EasySpot. He is responsible for day-to-day operations, revenue tracking, and ensuring good customer experience across all lots.

Motivation: Keep occupancy high and predictable while maximising revenue and minimising operational issues. António needs clear, timely information to justify decisions and report performance to stakeholders.

- Appreciates: Actionable key performance indicators (daily entries, occupancy rate, turnover), clear revenue figures and exportable reports, and fast alerts for issues that impact availability or customer satisfaction.
- Needs: A read-only dashboard showing daily entries, occupancy by zone and revenue summaries; visibility of fixed tariffs and historical billing for reconciliation; and a consolidated feed of customer reports and sensor alerts with full details to prioritise actions.

Persona #5 : The Maintenance Technician

Name: Laura Farias

Age: 29

Location: Alvalade, Portugal



Occupation: Maintenance Technician

Description: Laura Farias installs and maintains sensors and gateways for parking sites and expects full access to reported issues, sensor metadata and operational logs so she can diagnose faults and plan interventions.

Motivation: Laura wants complete, timely and precise information about sensor faults and customer reports so she can identify root causes remotely, prioritise repairs efficiently and minimise repeat visits and downtime.

- Appreciates: actionable key performance indicators (sensor uptime, false-positive rate, mean time to repair), detailed diagnostic logs and exportable reports, and immediate contextual alerts (sensor ID, last reading).
- Needs: a consolidated problems feed with sensor metadata; remote diagnostic tools (sensor's status); and the ability to create/assign prioritized maintenance orders with technical notes.

Scenarios

Scenario #1: Filipe Teixeira (regular driver)

Filipe is on his way to his weekly meeting in Lisbon with his clients. Unfortunately, due to unexpected traffic on the bridge, he is running late. He is worried that the time spent searching for a parking spot will further delay his meeting and damage his professional image. During the drive, he opens the EasySpot app to check the number of free spots and current rates at the parking lot, so he knows exactly what to look for when he arrives and how much it will cost him. If he feels that he might not be able to get a spot on time he can make a reservation within 30 minutes before arriving at the parking lot.

Scenario #2: Maria Silva (EV driver)

It is a Tuesday morning, and Maria is preparing for her 30-minute commute from Torres Vedras to her veterinary clinic. Her Tesla's battery is lower than usual, and she knows that finding a spot with a working charger near her office is crucial to avoid stress during her return trip.

Before leaving home, Maria opened the EasySpot app. She filters for available spots with EV charging stations within a 500-meter radius of her clinic. The app shows her a real-time map with a spot that includes a fast-charger compatible with her vehicle. She can see the current charging rates and the estimated occupancy for the next hour.

Scenario #3: Luís Pedro (driver with reduced mobility)

Luís has a follow-up medical appointment at a clinic. As a wheelchair user, he is often anxious about parking because standard spots do not provide enough space for him to exit his vehicle safely, and accessible spots are frequently occupied by unauthorized drivers.

Before leaving home, Luís uses the EasySpot app to filter for available accessible parking spaces closest to the clinic's entrance. By checking the app, he can see the exact distance he will need to travel in his wheelchair and confirm that the spots are being monitored, giving him the peace of mind that he will be able to attend his appointment on time and without physical strain.

Scenario #4: António Videira (Parking Manager)

António manages a network of parking lots and feels constantly frustrated by the lack of concrete data regarding his operations. The systems he currently uses are limited, offering superficial statistics that do not allow him to understand the true financial performance or the actual flow of customers in his assets.

Before starting his strategy meetings, António uses the EasySpot dashboard to gain a clear view of the business. By accessing the platform, he can monitor the exact number of customers who entered the lot daily and cross-reference that data with financial returns in real-time. Additionally, he validates the current tariffs applied and stays informed about all issues, ensuring that the operation runs flawlessly and that the investment is being profitable.

Scenario #5: Laura Farias (Maintenance Technician)

Laura is responsible for ensuring that the entire technological infrastructure of the park works perfectly. In her daily routine, she frequently faces the problem of receiving breakdown alerts that are too generic, which forces her to travel to the site without the right tools or without knowing exactly which sensor or device is failing.

Through EasySpot, Laura has access to a detailed technical panel where all reported problems appear with their specificities given by the users. She can consult the status of each sensor and the error history before even leaving the office.

Product requirements (User stories)

User Story #1: Real-time Parking Availability (High Priority)

As a regular driver,

I want to check the number of available parking spots in advance, so that I can minimize my search time and ensure I arrive at my meetings on time.

Given I am using the EasySpot app.

When I open the parking map

Then I should see the number of available spots updated in real time

User Story #2: Price Transparency & Planning (Low Priority)

As a regular driver,

I want to view the parking lot's rate structures and occupancy statistics, so that I can choose the most cost-effective timing for my stay and manage my expenses.

Given I am viewing a parking lot

When I open the pricing and occupancy section

Then I should see the hourly rates, tariffs, and historical occupancy trends

User Story #3: Parking Spot Reservation (Medium Priority)

As a regular driver,

I want to be able to make a reservation within the time limit (30 minutes before arriving)

Given I am driving to the parking lot

When I open the app on the number of available spots

Then I should be able to reserve a spot

And I should see how much it costs

User Story #4: Total of expenses spent on the parking lot (Low Priority)

As a regular driver,

I want to be able to see how much I have spent on the parking lot

Given I am a driver who parks at the parking lot

When I open the app on my expenses

Then I should see the total of my expenses

User Story #5: Report an unauthorized use of a parking spot (Low Priority)

As a client,

I want to be able to fill in a report if I see an unauthorized parking.

Given I am at a parking lot

When I select the option to report an issue

Then I should be able to submit a report

User Story #6: Real-time Charger Status (High Priority)

As an EV driver,

I want to see the real-time status and specification of EV chargers (speed, connector type), so that I can ensure my Tesla is compatible and a spot is available before I arrive.

Given I am searching for EV charging spots

When I open the EasySpot app and filter by EV spots

Then I should see its connector type, charging speed, and current status

User Story #7: Combined Parking + Charging Fee Overview (Low Priority)

As an EV driver,

I want to view combined parking and charging fees so that I can manage my daily commuting expenses and pay everything in a single transaction.

Given I am viewing an EV-enabled parking spot

When I open the pricing details

Then I should see the total estimated cost (parking + charging)

User Story #8: Accessible Spot Filtering & Proximity (High Priority)

As Mobility-Impaired Driver,

I want to filter the parking spots for the specifically designated accessible and see the exact distance to the parking entrance, so that I can ensure the walk the least distance to my medical appointment is manageable and within my physical limits.

Given I am using the app

When I apply the “Accessible Spots” filter

Then I should only see designated accessible spots

And I should see the distance to the parking entrance

User Story #9: Reliability & Physical Compliance (High Priority)

As a Mobility-Impaired driver,

I want to receive real-time confirmation of spot availability and space dimensions, so that I don't risk arriving at a spot that is either illegally occupied by an unauthorized vehicle or too narrow for me to safely deploy my wheelchair.

Given I am viewing an accessible spot

When I open its details

Then I should see its width, lateral clearance, and real-time occupancy

User Story #10: Real-Time Occupancy & Financial Performance (Low Priority)

As a parking operations manager,

I want to access a dashboard that shows the daily number of customers and real-time financial returns, so that I can evaluate the park's profitability and make data-driven decisions to optimize my investment.

Given I am logged into the operator dashboard

When I access the analytics section

Then I should see daily entries, occupancy rate, and revenue in real time

User Story #11: Tariff Management & Issue Oversight (Low Priority)

As a parking operations manager,
I want to view the current tariff structures and a centralized log of all problems, so that I can ensure the pricing strategy is correct and maintain a high standard of service quality across my facilities.

Given I am in the management dashboard

When I open the tariffs section

Then I should see all current tariffs and their effective dates

And when I open the issues log

Then I should see all reported problems with timestamps and categories

User Story #12: Technical Specificity & Remote Diagnostics (Low Priority)

As a field maintenance technician,
Accessing sensor logs and connection data will allow me to troubleshoot issues remotely and be fully prepared before traveling to the site.

Given I am viewing the maintenance panel

When I select a reported issue

Then I should see the sensor ID

And I should see the full error history

User Story #13: Infrastructure Mapping & Status Monitoring (High Priority)

As a field maintenance technician,
I want to see a live map of all hardware components and their individual status, so that I can perform proactive maintenance and ensure the parking ecosystem remains fully operational for all users.

Given I am in the technical dashboard

When I open the infrastructure map

Then I should see all sensors and their information

And each component should display its real-time operational status

User Story #14: Repair & Update of sensors (High Priority)

As a field maintenance technician,

I want to be able to update the state of the sensors when I repair a malfunction. So that the system accurately reflects their current operational condition.

Given I am on the technical dashboard

When I repair a sensor malfunction

Then I should be able to update the sensor's status

And the error statistics should be updated accordingly to reflect the change.